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Successive Overelaxation Method (SOR)

(for iterative solution to systems)

Like Jacobi (variant of Gauss-Seidel) but with an acceleration parameter.

Split in Gauss-Seidel

$$A = \underset{\text{lower}}{L} + \underset{\text{strictly upper}}{U}$$

(or)

$$A = \underset{\text{diagonal}}{D} + \underset{\text{strictly}}{L} + \underset{\text{strict}}{U}$$

<u>Now</u>:

$$\omega A = \underbrace{D + \omega L}_{M} + \underbrace{\omega U + (\omega - 1) D}_{N}$$

Iteration matrix

$$L_\omega := (D + \omega L)^{-1} (\omega U + (\omega - 1) D)$$

So

$$\omega A x = \omega b$$

$$\Rightarrow \quad (D + \omega L + \omega U + (\omega - 1) D) x = \omega b$$

$$\Rightarrow \quad (D + \omega L) x = \omega b - [\omega U + (\omega - 1) D] x$$

So
$$\boxed{\begin{gathered} x^{(k+1)} = -L_\omega x^{(k)} + \omega b \\ k = 0, 1, 2, \ldots \end{gathered}} \quad \text{(SOR)}$$

Convergence depends on
$$\rho(-L_\omega) = \rho(L_\omega)$$

Young is the inventor of this method.
Coincides with Gauss-Seidel for $\omega = 1$.

Results

- **Stein-Rosenberg Theorem**

  If $-L_\omega$ has nonnegative diagonal entries, then
  $$0 < \rho(L_\omega) < \rho(L_1) < 1$$
  for all $1 < \omega < 2$.

- **Ostrowski Theorem**

  For diag. dominant & positive definite matrices
  $$0 < \rho(L_\omega) < 1 \quad \text{for all } 1 < \omega < 2$$

- The optimal ω for SOR can be computed for special classes of matrices (d.d., p.d).

Make you likes hard:

Solve $\begin{bmatrix} 1.1 & 3.2 & -6 \\ 0.1 & 2 & 3 \\ 0 & 1 & 7.1 \end{bmatrix} x = \begin{bmatrix} 1.3 \\ 0 \\ -1 \end{bmatrix}$

by Jacobi, Gauss-Seidel and SOR using $\omega = 1.$

Compare number of iterations to reach a desired accuracy (say 10^{-3}).

Use/explore 2-3 starting vectors.

" Iteration is human, recursion is divine "

CHAPTER 2 Sensitivity of Linear Systems; Effects of Round-off Errors

§ 2.1 Vector & Matrix Norms

(now in $\mathbb{R}^n$ or $\mathbb{C}^n$)

Definition A (vector) norm on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ that assigns to every $x \in \mathbb{R}^n$ a nonnegative real number, $\|x\|$, such that the following hold:

(1) $\|x\| > 0$ if $x \neq 0$ & $\|x\| = 0$ iff $x = 0$

(2) $\|\alpha x\| = |\alpha| \, \|x\|$, $\forall \alpha \in \mathbb{R}$

(3) $\|x+y\| \leq \|x\| + \|y\|$, $\forall x, y \in \mathbb{R}^n$
(triangle inequality)

Notes

(i) Notice $\|0\| = \|0 \cdot x\|$ (for any x)
$\overset{(2)}{=} 0 \cdot \|x\| = 0$

(ii) $\|x-y\|$ is defined as the distance of $x, y \in \mathbb{R}^n$

<u>Example</u> Euclidean norm

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \text{ and } \|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2\right)^{1/2}$$

<u>Cauchy-Schwartz (CS) inequality</u>

$$\forall x, y \in \mathbb{R}^n : \left|\sum_{i=1}^{n} x_i y_i\right| \le \|x\|_2 \|y\|_2$$

<u>Proof</u> $\forall t \in \mathbb{R}$

$$\sum_{i=1}^{n} (x_i + t y_i)^2 = \underbrace{\sum x_i^2}_{c} + \underbrace{2t \sum x_i y_i}_{b} + \underbrace{t^2 \sum y_i^2}_{a}$$

$$= c + bt + at^2 \ge 0, \quad \forall t \ge 0$$

so quadratic ~~must~~ cannot have 2 distinct real roots, so

$$b^2 - 4ac \le 0$$

i.e. $\frac{b^2}{4} < ac$

i.e. $\left|\frac{b}{2}\right| < \sqrt{a}\sqrt{c} \rightsquigarrow$ CS ✓ ∎

<u>Triangle inequality</u> for $\|x\|_2$ (to prove it's a norm)

<u>Theorem</u> $\|x+y\|_2 \le \|x\|_2 + \|y\|_2$ (homework)

(Use CS !)

Other vector norms:

- 1-norm $\|x\|_1 = \sum_{i=1}^{n} |x_i|$
- ∞-norm $\|x\|_\infty = \max_{1 \le i \le n} \{|x_i|\}$
- Given any positive definite $A \in M_n(\mathbb{R})$ the A-norm on $\mathbb{R}^n$ is

$$\|x\|_A = (x^T A x)^{1/2}$$

prove it's a norm!

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Norms continued

Matrix norms

Definition Matrix norm

$$\|\cdot\| : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

such that $\forall A, B \in M_n(\mathbb{R})$, $\forall a \in \mathbb{R}$ we have

- $\|A\| > 0$ if $A \neq 0$
- $\|aA\| = |a|\,\|A\|$
- $\|A+B\| \leq \|A\| + \|B\|$ (triangle)
- $\|AB\| \leq \|A\|\,\|B\|$ (submultiplicativity)

Examples

(1) Frobenius norm: $A = [a_{ij}] \in M_n(\mathbb{R})$

$$\|A\|_F := \left(\sum_{i=1}^{n}\sum_{j=1}^{n} |a_{ij}|^2\right)^{1/2}$$

$$= \left(\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2\right)^{1/2}.$$

- Proof of triangle is analogous to Euclidean vector norm.
- Proof of submultiplicativity:

$$\|AB\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\sum_{k=1}^{n} a_{ik} b_{kj}\right)^2$$

$$\leq \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\sum_{k=1}^{n}|a_{ik}|^2 \sum_{k=1}^{n}|b_{kj}|^2\right)$$

(by applying Cauchy-Schwartz to $\left|\sum a_{ik} b_{kj}\right|$)

$$\leq \|A\|_F^2 \|B\|_F^2$$

(2) Every vector norm can be used to define a so-called "induced matrix norm"

Given $\|x\|_v$ vector norm, define

$$\|A\|_M := \max_{x \neq 0} \frac{\|Ax\|_v}{\|x\|_v}$$

Why is such a norm submultiplicative?
Proof in two steps:

<u>Theorem</u> A vector norm & its induced matrix norm satisfy

$$\forall x \in \mathbb{R}^n, \quad \|Ax\| \leq \|A\| \, \|x\|$$

Moreover, $\exists\, x \neq 0$ such that $\|Ax\| = \|A\| \|x\|$

To prove this theorem, let's prove C-S / triangle in inner product spaces over $\mathbb{C}$.

Define $\langle x, y \rangle = y^* x = \sum_{j=1}^{n} x_j \bar{y}_j$, $x, y \in \mathbb{C}^n$

"inner product"

$$\langle x, x \rangle = \sum_{j=1}^{n} x_j \bar{x}_j = \sum_{j=1}^{n} |x_j|^2 = \|x\|_2^2$$

($\|x\|_2$ = Euclidean norm of x)

<u>Theorem</u> (Cauchy-Schwartz)

$$\forall x, y \in \mathbb{C}^n, \ |\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$$

<u>Proof</u> (quick / tricky)

$\forall c \in \mathbb{C}$,

$$0 \leq \|x - cy\|_2^2 = \langle x - cy, x - cy \rangle$$

$$= \langle x, x - cy \rangle - \langle cy, x - cy \rangle$$

$$= \langle x, x \rangle - \bar{c} \langle x, y \rangle - c \langle y, x \rangle + c\bar{c} \langle y, y \rangle$$

Now $c = \dfrac{\langle x, y\rangle}{\langle y, y\rangle}$ $(y \neq 0)$ ← otherwise CS holds trivially.

Then

$$0 \le \langle x, x\rangle - \cancel{\frac{\overline{\langle x, y\rangle}}{\langle y, y\rangle}\langle x, y\rangle}$$

$$- \frac{\langle x, y\rangle}{\langle y, y\rangle}\langle y, x\rangle$$

$$+ \cancel{\frac{\langle x, y\rangle}{\langle y, y\rangle}\frac{\overline{\langle x, y\rangle}}{\cancel{\langle y, y\rangle}}\cancel{\langle y, y\rangle}}$$

$$\Rightarrow \quad 0 \le \langle x, x\rangle - \frac{\langle x, y\rangle}{\langle y, y\rangle}\langle y, x\rangle$$

$$= \langle x, x\rangle - \frac{\langle x, y\rangle\overline{\langle x, y\rangle}}{\langle y, y\rangle}$$

$$= \langle x, x\rangle - \frac{|\langle x, y\rangle|^2}{\langle y, y\rangle}$$

$$0 \le \|x\|_2^2 - \frac{|\langle x, y\rangle|^2}{\|y\|_2^2} \quad \blacksquare$$

<u>Theorem</u> $\forall x, y \in \mathbb{C}^n : \|x+y\|_2 \leq \|x\|_2 + \|y\|_2$.

<u>Proof</u>

$$\|x+y\|_2^2 = \langle x+y, x+y \rangle$$

$$= \langle x, x \rangle + \langle x, y \rangle + \overbrace{\langle y, x \rangle}^{\overline{\langle x, y \rangle}} + \langle y, y \rangle$$

$$= \|x\|_2^2 + 2\,\mathrm{Re}\langle x, y \rangle + \|y\|_2^2$$

$$\leq \|x\|_2^2 + 2\left|\mathrm{Re}\langle x, y \rangle\right| + \|y\|_2^2$$

(<u>But</u> $|z|^2 = (\mathrm{Re}\, z)^2 + (\mathrm{Im}\, z)^2 \geq (\mathrm{Re}\, z)^2$

$\Rightarrow |z| \geq |\mathrm{Re}\, z|$)

$$\leq \|x\|_2^2 + 2\left|\langle x, y \rangle\right| + \|y\|_2^2$$

C-S

$$\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

$$= (\|x\|_2 + \|y\|_2)^2$$

$$\Rightarrow \|x+y\|_2 \leq \|x\|_2 + \|y\|_2 \quad \blacksquare$$

Back to proving

$$\boxed{\|Ax\| \le \|A\| \, \|x\|}$$

<u>Proof</u> If $x=0$, trivial inequality!

Say $x \neq 0$. Then

$$\frac{\|Ax\|}{\|x\|} \le \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} =: \|A\|$$

Thus $\boxed{\|Ax\| \le \|A\| \, \|x\|.}$

"Moreover part"

The map $0 \neq x \xrightarrow{f} \frac{\|Ax\|}{\|x\|}$,

f is continuous & bounded.

The image of f is a closed bounded set so f attains its maximum

$$\exists \, \hat{x} \neq 0 : \frac{\|A\hat{x}\|}{\|A\| \, \|\hat{x}\|} = 1$$

$\frac{\|Ax\|}{\|A\| \cdot \|x\|} \le 1$

i.e. $\|A\hat{x}\| = \|A\| \, \|\hat{x}\|$ ∎

Why is $\|A\| := \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$ a <u>matrix norm</u>?

need proof.

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Move on matrix norms:

<u>Theorem</u> Every induced matrix norm satisfies the conditions of a matrix norm

$$\left(\|A\| := \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} \right)$$

<u>Proof</u>

- $\|A\| \geq 0$. If $A \neq 0$, $\exists \hat{x} : A\hat{x} \neq 0$

  Then $\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} \geq \frac{\|A\hat{x}\|}{\|\hat{x}\|} > 0$

  $(\|A\| = 0 \iff A = 0)$

- $\forall \alpha \in \mathbb{C}$: $\|\alpha A\| = \max_{x \neq 0} \frac{\|\alpha A x\|}{\|x\|}$

  $= |\alpha| \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = |\alpha| \|A\|$

- $\|A+B\| = \max_{x \neq 0} \frac{\|(A+B)x\|}{\|x\|}$

  $= \max_{x \neq 0} \frac{\|Ax + Bx\|}{\|x\|}$

  $\leq \max_{x \neq 0} \frac{\|Ax\| + \|Bx\|}{\|x\|}$

  $= \|A\| + \|B\|$

– Submultiplicativity:

$$\|AB\| = \max_{x \neq 0} \frac{\|ABx\|}{\|x\|}$$

$$\leq \max_{x \neq 0} \frac{\|A\| \, \|Bx\|}{\|x\|}$$

thm we proved

$$\leq \max_{x \neq 0} \frac{\|A\| \, \|B\| \, \cancel{\|x\|}}{\cancel{\|x\|}}$$

reapply same thm

$$= \|A\| \, \|B\| \qquad \blacksquare$$

Some matrix norms in use:

- $\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}$, $\quad 1 \leq p \leq \infty$

  $\left(\|x\|_p = \left(\sum_{j=1}^{n} |x_j|^p \right)^{1/p} \quad 1 \leq p < \infty \right)$

  $x = [x_1 \; x_2 \; \ldots \; x_n]^T \in \mathbb{C}^n$.

- $p = 2$: $\|A\|_2$ is known as the <u>spectral norms</u>

  (because $\|A\|_2$ can be computed via the Singular Value Decomposition of A which involve the eigenvalues of A^*A)

  ($\|A\|_2 = \sigma_1 = \sqrt{\lambda_1(A^*A)}$: largest singular value.

- All induced norms satisfy

$$\|I\| = 1$$

identity

because

$$\|I\| = \max_{x \neq 0} \frac{\|Ix\|}{\|x\|} = \max_{x \neq 0} \frac{\|x\|}{\|x\|} = 1$$

- Recall Frobenius norm of $A = [a_{ij}] \in M_n(\mathbb{C})$

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2}$$

Here $\|I\|_F = \sqrt{n} > 1$ if $n > 1$.

Thus the Frobenius norm is <u>not</u> an induced norm.

- $\|A\|_2 \leq \|A\|_F$ (use C-S to prove)

- $\|A\|_1 = \max_j \sum_{i=1}^{n} |a_{ij}| = \max$ "column" sum

$\|A\|_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| = \max$ "row" sum.

(These two are harder to prove)

§2.2 Condition numbers

Goal: measure sensitivity of solving $Ax=b$ under perturbations of b (to start with)

Basic assumptions
- $b \neq 0$
- A is invertible ($Ax=b$ has unique solution)

Suppose instead of $Ax=b$ we solve:

$$A\hat{x} = b + \delta b ,$$

$\hat{x}$: notation for solution to perturbed system

$A \in M_n(\mathbb{C})$, $b, \delta b \in \mathbb{C}^n$ (δb: no assumptions)

Let $\delta x := \hat{x} - x$ (x: actual)

i.e.

$$A(\underbrace{x + \delta x}_{\hat{x}}) = Ax + A\delta_x = b + \delta b$$

$$\Rightarrow \quad A\delta x = \delta b$$

$$\Rightarrow \quad \delta x = A^{-1}\delta b$$

$$\Rightarrow \boxed{\|\delta x\| = \|A^{-1}\delta b\| \le \|A^{-1}\|\,\|\delta b\|} \quad (1)$$

Similarly $\|b\| = \|Ax\| \le \|A\|\,\|x\|$

$$\Rightarrow \boxed{\frac{1}{\|x\|} \le \|A\| \frac{1}{\|b\|}} \quad (2) \qquad (b \neq 0 \Rightarrow x \neq 0)$$

Thus by (1) & (2)

$$\boxed{\frac{\|\delta x\|}{\|x\|} \le \|A\|\,\|A^{-1}\| \frac{\|\delta b\|}{\|b\|}} \quad (*)$$

rel error in the computed solution

relative error in perturbed r.h.s.

The error is controlled largely by

$$\kappa(A) := \|A\|\,\|A^{-1}\|$$

which is called the <u>condition number</u> of A

$\kappa(A) =$
- "large": dangerous; call A <u>ill-conditioned</u>
- "small": trust solutions; call A <u>well-conditioned</u>

<u>Proposition</u> For every induced norm:

(a) $\|I\| = 1$, $\kappa(I) = 1$

(b) $\kappa(A) \geq 1$, $\forall A \in M_n(\mathbb{C})$

<u>Proof</u> (a) $\kappa(I) = \|I\| \|I^{-1}\| = 1 \cdot 1 = 1$

(b) $\kappa(A) = \|A\| \|A^{-1}\| \geq \|AA^{-1}\|$

$= \|I\| = 1$

<u>Note</u> Obviously, condition number depends on the norm

$$\kappa_p(A) = \|A\|_p \|A^{-1}\|_p$$

(usually $p = 1, 2, \infty$)

<u>Example</u>

$$A = \begin{bmatrix} 1000 & 999 \\ 999 & 998 \end{bmatrix} \quad (\det A = 1)$$

$$\Rightarrow A^{-1} = -\begin{bmatrix} -998 & 999 \\ 999 & -1000 \end{bmatrix} \quad (\det A^{-1} = 1)$$

Let's solve $Ax = b$:

$$\begin{cases} 1000x_1 + 999x_2 = b_1 & : \text{line } 1 \\ 999x_1 + 998x_2 = b_2 & : \text{line } 2 \end{cases}$$

Slope of $l_1 = -\frac{1000}{999} = -1.001001$

$l_2 = -\frac{999}{998} = -1.001002$

These two ...